

Junsung Park

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Research Interests

Robotics Engineer | Learning-Based Control

I aim to **build generalizable robot autonomy capable of robust interaction in dynamic, unstructured environments**. My primary interest lies in **Robot Learning**, utilizing **Reinforcement Learning** and **Learning based Control** to ensure system robustness against uncertainty.

Ultimately, I seek to advance **Human-Robot Interaction**, focusing on humanoid robots and wearable interfaces to bridge the gap between human intent and robotic action.

Education

Carnegie Mellon University, Pittsburgh, PA, USA

Aug 2026 – May 2028

- M.S. in Mechanical Engineering (Research Track)

University of California, San Diego, La Jolla, CA, USA

Jan 2026 - Present

- Exchange Student
- Tuition and stipend fully covered by Korea-U.S. Student Exchange Program Scholarship

Seoul National University, Seoul, South Korea

Mar 2020 – Feb 2026

- B.S. in Department of Electrical and Computer Engineering (Minor in Mechanical Engineering)
- **GPA: 3.89/4.0**
- Relevant Coursework: Robot Learning (Graduate, A+), Mechanical System Modeling and Control (A+), Introduction to Robotics (A+), Fundamentals of Control Engineering (A+), Mechatronics (A+)
- Tuition and stipend fully covered by Silla Scholarship Foundation
- Leave of absence for 2 years of mandatory military service in the Republic of Korea Air Force

Daejeon Science High School for the Gifted, Daejeon, South Korea

Mar 2017 – Feb 2020

- Graduated with Honors, Specialized Curriculum in Mathematics and Physics

Publications

[1] **Junsung Park**. *BiCQL-ML: A Bi-Level Conservative Q-Learning Framework for Inverse Reinforcement Learning*. arXiv preprint arXiv:2511.22210, 2025. (link)

[2] **Junsung Park**, Hogun Kee, and Songhwai Oh. *Modality-Augmented Fine-Tuning of Foundation Robot Policies for Cross-Embodiment Manipulation on GR1 and G1*. arXiv preprint arXiv:2512.01358, 2025. (link) (*Working in Progress*)

[3] Anya Zorin, Zilin Si, Myungsun Park, **Junsung Park**, Alexiy Buynitsky, Sachin Bhadang, Taejun Park, Sohee John Yoon, Yong-Lae Park, Oliver Kroemer, Zeynep Temel, Michael T. Tolley, Sha Yi, and Xiaolong Wang. *TacO: Benchmarking Tactile Sensors for Object Manipulation*. Submitted to *IEEE Robotics: Science and Systems (RSS)*, 2026. (*Under Review*)

Research Experience

Wang Lab, UC San Diego, *Research Intern* (Advisor: **Prof. Xiaolong Wang**)

Jan 2026 – Present

- Co-developed **TacO**, a large-scale **tactile sensing benchmark** for robotic manipulation, evaluating diverse sensors across complex contact-rich tasks.
- Integrated **multi-modal tactile feedback** into reinforcement learning frameworks to enhance the robotic grasping and in-hand manipulation.

Robot Learning Lab (RL LAB), *Research Intern* (Advisor: **Prof. Songhwai Oh**)

Jan 2025 – Dec 2025

- Pioneered **modality-augmented VLA diffusion policies** for the **Unitree G1**, creating a **multi-modal dataset**

that improved manipulation success from 48% to 94% via force integration - **open-sourced the dataset on Hugging Face [Link]**, achieving over 4k downloads in the first month.

- Developed a **goal-conditioned locomotion controller** using **end-to-end PPO** with **curriculum reward shaping**, enabling stable navigation and robust heading alignment.

Soft Robotics and Bionics Lab, Research Intern (Advisor: **Prof. Yong Lae Park**)

June 2024 – Dec 2024

- Designed and prototyped a **wearable rehabilitation glove** integrating **Twisted String Actuators** and **e-gain soft sensor** for assisted hand motion.
- Developed and integrated motor control architecture for real-time finger strain detection, enhanced by **IMU-based feedback control** to enable adaptive and **closed-loop motion regulation**.

Conference Presentations

2025 IEEE MIT URTC, Boston, MA, USA

Oct 2025

- Poster Presentation (Peer-Reviewed Abstract): *“BiCQL-ML: A Bi-Level Conservative Q-Learning Framework for Inverse Reinforcement Learning.”*

2025 ICCE-ASIA, Busan, South Korea

Oct 2025

- Poster Presentation: *“BiCQL-ML: A Bi-Level Conservative Q-Learning Framework for Inverse Reinforcement Learning.”*

2024 SRRC Workshop, Seoul, South Korea

Feb 2025

- Design and Implementation of a TSA-Driven Opposition Assist Glove with Integrated Soft Sensors
- Presented a poster on the development of a wearable rehabilitation glove integrating rack-and-pinion actuator, e-gain soft sensor and IMU-based feedback control for adaptive hand motion assistance.

Awards and Honors

Gold Prize, SNU Creative Design Fair (out of 75 teams)

Sep 2025

- Awarded with a prize of \$1,000 and international conference support worth \$4,000
- Developed and validated a novel offline IRL reward-inference method through predictive-consistency evaluation on MuJoCo continuous-control benchmarks without online RL.

3rd Prize, Mechatronics Design Competition (out of 25 teams)

June 2024

- Awarded with a prize of \$500
- Recognized for developing an innovative mechatronic system integrating mechanical design and control algorithms

Korea-U.S. Student Exchange Program Scholarship

Nov 2025

- Recognized for excellence in intelligent robotics research; awarded \$9,000 for outstanding academic and interdisciplinary achievement in AI-driven communication and control systems.

Semiconductor Track Scholarship

Jan 2025 - Present

- Recognized for excellence in interdisciplinary studies; awarded \$4,300 to date, with \$10,000 to be awarded upon graduation

Shilla Cultural Scholarship Foundation Scholarship

Mar 2021 - Present

- Awarded full tuition coverage until graduation, recognizing academic excellence

Skills

Languages: Korean, English

Technologies: Python, C++, MATLAB, Isaac Sim, Fusion360, Simulink, Verilog